MIL-T-2764C(SHIPS)
13 March 1972
SUPERSEDING
MIL-T-2764B(SHIPS)
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(See 6.5)

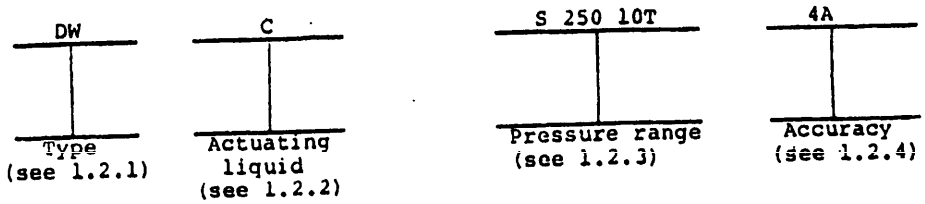
MILITARY SPECIFICATION

TESTERS, DEAD WEIGHT PISTON

1. SCOPE

1.1 Scope. This specification covers dead weight piston testers for use in test pressure instruments.

1.2 Classification. Testers shall be of a design having the following variables (see 6.1):



1.2.1 Type. Testers shall be the following type:

Symbol	Type
DW	Dead weight piston

1.2.2 Actuating liquid. Actuating liquid for testers shall be one of the following:

Symbol	Color code	Liquid
C	Blue	Water
D	Brown	Oil

1.2.3 Pressure range. Pressure ranges for testers shall be as follows:

Symbol	Range, pounds per square inch gage (p.s.i.g)
S 10 3M	Single range, 10 to 300
S 10 6M	Single range, 10 to 600
S 10 1T	Single range, 10 to 1,000
S 125 3T	Single range, 125 to 3,000
S 125 6T	Single range, 125 to 6,000
S 250 10T	Single range, 250 to 10,000
D 10 6T	Dual range, 10 to 6,000
D 10 10T	Dual range, 10 to 10,000
D 10 15T	Dual range, 10 to 15,000

1.2.4 Accuracy. Testers shall have an accuracy designation as follows:

Symbol	Limit of errors
4A	± 1/10 percent of reading

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids o request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585 - Boxes, Wood, Wirebound.

FSC 668

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FEDERAL (cont'd.)
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-P-40 - Packaging and Packing of Hand Tools.
PPP-T-60 - Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper (For Carton Sealing).

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-Q-9858 - Quality Program Requirements.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-P-15024 - Plates, Tags and Bands For Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-M-15071 - Manuals, Technical: Equipments and Systems Content Requirements For.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 11).
MIL-F-21467 - Fittings, Flareless, Fluid Connection (Shipboard Use).

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MS16142 - Boss, Gasket Seal Straight Thread Tube Fitting, Standard Dimensions For.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
NBS Monograph 65 - Reduction of Data For Piston Gage Pressure Measurements.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT
National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, Inc. Agent, 1616 P Street, N.W., Washington, D.C. 20036.)

3. REQUIREMENTS

3.1 Sample for first article inspection. When specified in the contract or order, prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.3).

3.2 Materials. Materials shall be as specified herein. When not specified, materials of the best commercial grade and suitable for the intended service shall be furnished.

3.2.1 Nonferrous metals. Nonferrous metal material shall be nickel-copper alloys.

3.2.2 Corrosion resisting metals. Metal parts required to be corrosion resisting metals shall be brass, bronze, copper-nickel, or corrosion resisting steel.

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3.2.2.1 Aluminum. Use of aluminum shall be limited to parts not exposed to the actuating liquids. Aluminum shall be anodized or acid dipped to prevent corrosion and protective coating added in accordance with 3.2.2.2.

3.2.2.2 Finish. Protective coating, where required, shall include an air dry primer with grey enamel finish in accordance with MIL-E-15090, formula 111. Machined parts, threaded parts and castings shall have burrs and chips removed. The finish of moving parts shall be suitable for the intended purpose of the part. Parts of corrosion resisting materials shall not require protective coatings.

3.2.3 Gaskets and seals. Nonmetallic gaskets and seals shall be suitable for automotive oil, distilled water and freon cleaning solvents.

3.2.4 Actuating liquid. The actuating liquid shall be one of the following:

Symbol	Color code	Liquid
C	Blue	Distilled, demineralized tap water.
D	Brown	Automotive oil (SAE-20)

3.3 Design. Testers shall be designed to be assembled as a test system and mounted on a flat level surface such as a bench top for use in calibrating pressure instruments. The test system shall include a hand-operated liquid pump, a piston-cylinder-platform subassembly with weights and means for connecting the test system to the instrument being testec The tester shall also include tools actuating liquid, and other accessories for dismantling the tester and carrying cases for use when transporting the tester to a laboratory for calibration verification.

3.3.1 Pressure ranges. Testers shall have single or dual ranges in accordance with table I. Dual range testers shall include a separate piston-cylinder-platform subassembly for the low pressure range and a separate subassembly for the high pressure range.

Table I - Pressure ranges.

Symbol	Pressure range and increments
	Single range testers
S 10 3H	10 p.s.i. or less to 300 p.s.i. in increments of not more than 5 p.s.i.
S 10 6H	10 p.s.i. or less to 600 p.s.i. in increments of not more than 5 p.s.i.
S 10 1T	10 p.s.i. or less to 1000 p.s.i. in increments of not more than 10 p.s.i.
S 125 3T	125 p.s.i. or less to 3000 p.s.i. in increments of not more than 25 p.s.i.
S 125 6T	125 p.s.i. or less to 6000 p.s.i. in increments of not more than 50 p.s.i.
S 250 10T	250 p.s.i. or less to 10,000 p.s.i. in increments of not more than 50 p.s.i.
	Dual range testers
D 10 6T	10 p.s.i. or less to 600 p.s.i. in increments of not more than 5 p.s.i. 100 p.s.i. or less to 6000 p.s.i. in increments of not more than 50 p.s.i.
D 10 10T	10 p.s.i. or less to 1000 p.s.i. in increments of not more than 5 p.s.i. 100 p.s.i. or less to 10,000 p.s.i. in increments of not more than 50 p.s.i.
D 10 15T	10 p.s.i. or less to 1500 p.s.i. in increments of not more than 5 p.s.i. 100 p.s.i. or less to 15,000 p.s.i. in increments of not more than 50 p.s.i.

3.3.2 Test system operation. With the test system mounted, assembled and filled with the actuating liquid, operating the pump shall increase the hydraulic pressure to the piston-cylinder-platform subassembly and to the instrument being tested. Test system pressure shall be balanced against the weights on the platform when rotating slowly to minimize friction and compared to the readings on the instrument being tested. Errors in the test system shall be determined in accordance with National Bureau of Standards (NBS) Monograph 6

3.3.3 Limits of errors. The test system, assembled and operated in the prescribed manner, shall have limits of errors not exceeding plus or minus 1/10 of 1 percent of indicate reading as determined in accordance with NBS Monograph 65 and by testing.

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3.4 Detail requirements.

3.4.1 Hand-operated liquid pump. The pump shall include a base, body, pump mechanism and a reservoir for the actuating liquid. The reservoir capacity shall be adequate for filling the entire test system, operate the system to maximum pressure and with not less than 10 percent reserve liquid without additional liquid. Means shall be provided for filling, draining and visually determining that the quantity of stored liquid is adequate. The system shall include filters, strainers or other devices to prevent wear or damage to critical parts from contaminates in the actuating liquid. The external surface of the pump shall have a protective coating in accordance with 3.2.2.2. Parts made of corrosion resisting material or nonferrous material shall not require a protective coating. A pump identification plate shall be furnished and shall include, maximum discharge pressure, model number, manufacturer's name, and serial number.

3.4.1.1 Pump operation. The applied force on the pump handle required to raise system pressure to any value up to and including 30 percent above the maximum range pressure shall not exceed 50 pounds. The pump action shall be smooth and shall maintain system pressure at any point in the pump stroke. The pump shall include means for micrometer or vernier adjustment of the system pressure. Maximum system pressure shall be achievable within 30 minutes when operated by one man. The pump shall be capable of operating from minimum range value to maximum range pressure 200 times without leakage, excessive wear, maintenance or failure.

3.4.1.2 Piping. Piping, fittings, plugs and other components exposed to pump discharge pressure shall be designed to be suitable for not less than maximum pump discharge pressure.

3.4.2 Piston-cylinder-platform subassembly. The piston-cylinder-platform subassembly shall be of corrosion resisting materials designed to maintain uniform dimensions at all pressures up to the maximum range value without leakage. Piston and cylinders shall be fitted and shall have area accuracy of plus or minus 1/20 of 1 percent or greater. Inlet and outlet shall be marked. The platform shall be marked with the equivalent weight.

3.4.3 Weights. The weights shall be of nonmagnetic, solid, nonporous, corrosion resisting materials. The weights shall be cylindrical in shape, recessed for nesting and self-aligning when stacked on the platform. Weights of equal rated value shall be of a uniform shape and each weight, including the platform, shall be accurate to plus or minus 1/20 of 1 percent or greater. Quantities of each size of weights furnished shall provide step changes in system pressures of the smallest increment from minimum range value to the maximum range pressure. Weights shall be marked with equivalent pressure.

3.4.3.1 Operation. The piston-cylinder-platform subassembly when leveled and stacked with weights equal to maximum range value, pressurized and rotated slowly to reduce friction, shall be stable and designed to minimize wear of the piston and cylinder due to corkscrewing or eccentric loading.

3.4.4 Accessories. Each tester shall include the following accessories:

(a) Tools. A set of tools required to assemble and disassemble the test system for storing in the carrying cases and for maintenance of the equipment. Tools shall be machined, tempered or blued steel.

(b) Adaptors. Adaptors for testers being tested shall be of corrosion resisting materials with designs of connections suitable to assure leak tight joints after frequent take-downs. Adaptors shall include the following:

Taper pipe threads (one each)

1/8 inch male NPT	1/8 inch female NPT
1/4 inch male NPT	1/4 inch female NPT
3/8 inch male NPT	3/8 inch female NPT
1/2 inch male NPT	1/2 inch female NPT

MS16142 (straight thread female "O" ring seal boss)

7/16 - 20NS

MIL-F-21467 (flareless, male, straight threads)

7/16 - 20NS

Note: Adaptors shall be furnished in a container labeled "ADAPTOR KIT" for storage in the carrying case.

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(c) Spare seals and gaskets. A set of spare seals and gaskets shall be furnished which includes a replacement unit individually packaged, to provide maximum shelf life for each seal and gasket used in the tester.

(d) Manual. An operation, maintenance and parts list manual shall be furnished with each tester. The manual shall be as specified in 3.5.

(e) Liquid charge. Liquid charge containers shall be unbreakable plastic, reuseable with leak tight caps and shall include dispensing tubes. Each container shall be sized for a full charge and shall be permanently marked "Hydraulic Oil" or "Distilled Water".

3.4.5 Carrying cases. Dead weight piston testers shall be furnished with metal carrying cases for the pump, accessories, and piston-cylinder-platform subassembly and separate cases with individual compartments for the weights. Each stored part shall have a designated location and retainers to prevent displacement. Each case shall have a cover, handles and latches which shall not disengage when the case is being carried. The weight of each case shall not exceed 125 pounds. Carrying cases shall have a protective coating in accordance with 3.2.2.2.

3.4.5.1 Identification plate. Each carrying case shall have an identification plate attached to the external surface on the top. The identification plate shall be metal materia and in accordance with MIL-P-15024 and MIL-P-15024/5.

3.4.5.1.1 Pump and accessories carrying case. The identification plate shall have a background color to identify the actuating liquid as follows:

Color	Actuating liquid
Blue	Water
Brown	Oil

The following information shall be included:

DEAD WEIGHT TESTER

PUMP AND ACCESSORIES

ACTUATING LIQUID _____

Model Number _____
Manufacturer _____
Serial Number 1/ _____
Federal Stock Number 1/ _____

3.4.5.2 Weights carrying case. The identification plate shall include the following information:

DEAD WEIGHT TESTER

WEIGHTS

Model Number _____
Manufacturer _____
Range _____ p.s.i.g. min. _____ p.s.i.g. max. _____
Serial Number 1/ _____
Federal Stock Number 1/ _____

3.4.5.3 Information plate. A metal information plate in accordance with MIL-P-15024/5 shall be furnished, attached to the inside of the cover of the pump and accessories carrying case and shall include test system assembling instructions.

3.5 Manuals. Two copies of the operation and maintenance manual shall be furnished with each tester stored in the pump and accessories carrying case. The manual shall be type I of MIL-M-15071.

1/ Serial or Federal stock number shall be the same for all cases making up the tester.

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3.6 Workmanship. Workmanship shall be of a quality to provide a finished tester having parts and accessories free from burrs, scratches, chips, defects, corrosion, or machining products or other conditions which may affect appearance, mating of parts, dimensional accuracy and performance in accordance with this specification.

3.6.1 Cleaning. All parts, assemblies and subassemblies shall be cleaned and free from foreign material which may cause pressure connections to leak, defective pump operation, or early failure of wearing parts.

3.6.2 Fastening devices. Screws, nuts, bolts and other fastening devices shall be free from defects which may cause the fastener to improperly loosen during operation, prevent proper assembling, or adjustment. When subject to removal or adjustment, fasteners shall not be swaged, peened, staked or otherwise permanently deformed.

3.6.3 Castings. Castings shall be uniform, free from blow holes, porosity, hard spots, shrinkage, cracks or other defects affecting suitability of the casting for the intended purpose.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Quality program. The supplier shall provide and maintain a quality program acceptable to the Government for supplies and services covered by this specification. The quality program shall be in accordance with MIL-Q-9858 (see 6.1 and 6.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. A sample tester having the highest pressure range and including all parts required for a complete tester shall be subjected to the examination and tests of table II.

4.3.1 First article inspection report. A report of the first article inspection shall be submitted to the Naval Ship Engineering Center (NAVSEC) for approval.

4.4 Quality conformance inspection.

4.4.1 Lot. For purposes of inspection, a lot shall consist of all testers offered for delivery at one time.

4.4.2 Sampling for quality conformance inspection. A random sample of testers shall be selected from each lot in accordance with MIL-STD-105 at Inspection Level II and Acceptable Quality Level (AQL) of 2.5 percent defective, for the examination and tests of table II.

Table II - Inspection.

Inspection	Inspection schedule		Requirement paragraph	Test paragraph
	First article inspection	Quality conformance inspection		
Examination	X	X	3.3	4.5
Test system operation	X	--	3.3.2	4.6.1.1
Errors determination:				
Piston-cylinder	X	X	3.4.2	4.6.2.1
Weights	X	X	3.4.3	4.6.2.2
System	X	--	3.3.3	4.6.2.3
Pump:				
Pressure	X	X	3.4.1.1	4.6.3.1
Operation	X	--	3.4.1.1	4.6.3.1
Life cycle	X	--	3.4.1.1	4.6.3.2
Hydrostatic, piping and fittings	X	--	3.4.1.2	4.6.4

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4.5 Examination. The testers selected in accordance with 4.4.2 shall be examined to determine conformance to the requirements of this specification and the classification of defects in table III.

Table III - Classification of defects.

Categories	Defects
Critical:	
1	Errors of weights excessive.
2	Errors of piston-cylinder-platform excessive.
3	Pump incapable of maximum pressure.
4	Test standards not traceable to National Standards; not checked within required frequency; data history nonconforming or not available.
Major:	
101	Pump requires greater than 50 pounds force on the pump handle or greater than 30 minutes to achieve maximum range pressure.
102	Weights unstable on platform when rotated slowly, will not stack, tilts, falls over; finish scratched, dented or damaged.
103	Piston and cylinder surfaces not polished, honed or fitted; scratched; leaks under pressure.
104	System leaks under pressure; pressure connections not suitable for hydrostatic pressure test; threads nonconforming and machined edge rough, not smoothed, burrs not removed; sharp; wrong thread size; stripped or defective.
105	Pressure fittings not suitable for maximum range pressure with zero leakage after frequent disassembly; gaskets and seals wrong material, require replacement frequently.
106	Labels nonconforming, not properly marked, not furnished, not metallic, not properly attached, weights not marked, incorrect data furnished; pump data incomplete, not furnished; piston-cylinder-platform not marked with equivalent platform value; assembly procedures not included on plate in carrying case.
107	Carrying cases nonconforming, not metal materials, weight limit exceeded when loaded with tester parts and accessories, carrying case handles, latches and retainers not suitable or not furnished, inadequate when being carried; storage space inadequate.
108	Accessories, nonconforming, not furnished; tools not machined or blued steel, wrong size, inadequate for assembling tester; manual not furnished, incomplete; adaptors not furnished, incomplete set, not required size, liquid charge not furnished, not required type.
109	Materials not otherwise covered, nonconforming, finish scratched unacceptable.
Minor:	
201	Paint finish nonconforming, scratched, wrong color.

4.6 Test procedures.

4.6.1 Assembling the tester. A complete tester shall be assembled and filled using the accessories furnished with the tester and in accordance with the instruction manual.

4.6.1.1 Tester operation test. The assembled test system shall be operated to maximum range pressure two times. During the system test all operating features and performance requirements shall be observed and verified.

4.6.2 Determination of errors.

4.6.2.1 Piston and cylinders. Dimensions of piston and mating cylinder shall be measured and tabulated with the temperatures recorded during tabulation. Data recorded shall be adequate for data reduction in accordance with NBS Monograph 65, sections 4 and 5.

4.6.2.2 Weights. Mass of weights and the platform shall be measured and tabulated with the temperatures, and taking into account correction factors of NBS Monograph 65, section 3.

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4.6.2.3 System. The collected data shall be analyzed to show that errors of the system do not exceed plus or minus 1/10 of 1 percent of reading in accordance with sections 8, 9 and 10 of NBS Monograph 65.

4.6.3 Pump tests.

4.6.3.1 Pressure and operation test. The pump shall be tested separate from the test system. A test gage suitable for maximum pump discharge pressure shall be connected to the pump outlet. The pump shall be filled with the liquid charge and gas vented from the pressure system. The pump shall be operated to the maximum pressure. Force on the pump handle shall be determined at maximum range pressure. It shall be verified that maximum range pressure can be achieved in 30 minutes. Test shall be performed without failure, maintenance or replacement of parts.

4.6.3.2 Life cycle test. The pump shall be operated to maximum range pressure 200 times. This shall be accomplished without failure, maintenance, or replacement of parts except those furnished with the tester.

4.6.4 Hydrostatic test. Each of the testers shall be hydrostatically tested to determine compliance with 3.4.1.2.

4.7 Inspection of preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The equipment shall be cleaned by process C-1 of MIL-P-116. Equipment shall be dried in accordance with any drying procedure specified in MIL-P-116.

5.1.1.2 Testers. Immediately after cleaning and drying, the interior surfaces of each tester shall be flushed with preservative compound conforming to P-9 of MIL-P-116 and thoroughly drained of all excess preservative compound. All exterior ferrous surfaces that are not painted, plated or treated to resist corrosion shall be coated with preservative compound conforming to P-2 of MIL-P-116. Exterior surfaces coated with a preservative compound shall be enclosed in a greaseproof wrap and secured in accordance with MIL-P-116. Painted surfaces on which paint film has become damaged shall have the damaged surfaces cleaned and repainted as specified in 3.2.2.2.

5.1.1.3 Accessories and weights. Accessories and weights shall be preserved and packaged in accordance with the guide lines of MIL-P-116.

5.1.1.4 Tools. Tools shall be preserved and packaged in accordance with PPP-P-40.

5.1.1.5 Intermediate packaging. The tester, accessories, tools and technical manuals packaged as specified and comprising one complete unit shall be placed and secured in the case provided. Weights shall be placed in the same carrying case with tester or in the carrying case provided. Equipment shall be so arranged, separated and cushioned so as to prevent damage from movement during handling, shipment, and storage. The carrying case lids shall be closed and secured with the fastening devices provided. The lid joint shall be sealed with pressure sensitive tape conforming to type I or II, class 1 of PPP-T-60.

5.1.1.6 Unit containers. Each tester case, (see 3.4.5 and 5.1.1.5), shall be placed in a fiberboard box conforming to PPP-B-636. Containers shall be closed and reinforced in accordance with the appendix to the box specification. Closure shall be accomplished by use of pressure-sensitive tape. When specified (see 6.1) unit containers may be used as the shipping container.

5.1.2 Level C.

5.1.2.1 Preservation-packaging. Preservation-packaging shall afford protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early use.

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5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Unless otherwise specified (see 6.1), testers, preserved-packaged as specified shall be packed in containers conforming to any one of the following specifications at the option of the supplier:

Specification	Container	Type or class
PPP-B-576	Wood-cleated veneer, paper overlaid	Overseas, class 2
PPP-B-585	Wood-wirebound	Class III
PPP-B-591	Fiberboard, wood cleated	Class II
PPP-B-601	Wood, cleated-plywood	Overseas
PPP-B-621	Wood, nailed and lock corner	Overseas, class 2
PPP-B-636	Fiberboard	Weather-resistant
PPP-B-640	Fiberboard-corrugated triple wall	Class 2

Shipping containers shall be closed, strapped or reinforced and waterproofed in accordance with the applicable container specification or appendix thereto and as specified herein. Unless otherwise specified (see 6.1), shipping containers shall have case liners conforming to the requirements of MIL-L-10547. Case liners for fiberboard containers conforming to PPP-B-636 weather-resistant and PPP-B-640 class 2 may be omitted provided all center and edge seams and the manufacturer's joint of the container are sealed with minimum 2-inch wide tape conforming to PPP-T-60, class 1 or PPP-T-76. Fiberboard boxes shall be reinforced with pressure-sensitive, reinforced, filament tape applied in accordance with the appendix to the applicable fiberboard box specification, in lieu of steel strapping. Unit containers conforming to PPP-B-636, class weather-resistant, when closed, sealed and reinforced as specified herein and used as a shipping container need not be over-packed. The gross weight of wood, wood-cleated, or triple-wall fiberboard boxes shall not exceed 200 pounds. Fiberboard containers, PPP-B-636 shall not exceed the weight limitations of the container specification.

5.2.2 Level B. Unless otherwise specified (see 6.1), testers, preserved-packaged as specified shall be packed in containers conforming to any one of the following specifications at the option of the supplier:

Specification	Container	Type or class
PPP-B-576	Wood-cleated veneer, paper overlaid	Class 1
PPP-B-585	Wood-wirebound	Class II
PPP-B-591	Fiberboard, wood cleated	Class 1
PPP-B-601	Wood, cleated-plywood	Domestic
PPP-B-621	Wood, nailed and lock corner	Domestic, class 1
PPP-B-636	Fiberboard	Domestic
PPP-B-640	Fiberboard-corrugated triple wall	Class 1

Shipping containers shall be closed, strapped or reinforced in accordance with the applicable container specification or appendix thereto. Fiberboard containers, PPP-B-636 and PPP-B-640, shall be closed with pressure-sensitive tape. Unit containers conforming to PPP-B-636, closed and reinforced as specified herein and used as shipping containers, need not be over-packed. The gross weight of wood, wood-cleated or triple wall fiberboard boxes shall not exceed 200 pounds. Fiberboard containers shall not exceed the weight limitations of the applicable container specification.

5.2.3 Level C. Testers, preserved-packaged as specified shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing, or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules or other carrier rules as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.1), interior and exterior shipping containers shall be marked for shipment in accordance with MIL-STD-129.

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5.4 Use of polystyrene (loose-fill) material. Unless approved by the procuring activity (see 6.1), use of polystyrene "loose-fill" material for packaging or packing applications such as cushioning, filler, dunnage, etc., is prohibited. When approved, container(s) (interior and exterior) shall be marked or labelled as follows:

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken onboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard or transparent flexible cellular material."

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, actuating liquid, pressure range, and accuracy (see 1.2).
(c) Quality assurance provisions (see 4.1.1).
(d) Level of preservation-packaging and level of packing required (see 5.1 and 5.2).
(e) When unit containers may be used as the shipping container (see 5.1.1.6).
(f) Shipping container, if other than supplier's option (see 5.2.1 and 5.2.2).
(g) Special marking (see 5.3 and 5.4).
(h) When polystyrene "loose-fill" material is approved (see 5.4).

6.2 Management control system document. The following management control system document should be included on DD Form 1660:

(a) MIL-Q-9858 (see 4.1.1).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE WERE MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

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